Landau's theoretical minimum: Paper 004

Companion for v0.1.0. Read the student guide, solve these questions, then consult the separate instructor key. All Apartment 50 dialogue is fictional.

Questions 1–6 establish vocabulary; 7–14 require calculations or precise contract decisions; 15–18 are the final challenge round. Work over exact integers/rationals unless a question explicitly names bytes. A correct explanation of unavailable evidence is a valid answer; silently replacing the problem is not.

Round 1: the receipt has arrived

1. Behemoth compares `ab` and `ba` and announces that equal lengths mean equal digests. Identify the mistake. If two artifacts match a retained expected digest, what does that establish, and what does it leave open?
2. A proposal has zero final Phi1 and Gamma2, partial conformance, and unsigned laboratory attestation. Give its execution label, admission verdict and two admission codes. Which part is a mathematical result and which part is a policy decision?
3. An integer `7` and string `"7"` pass through the domain canonical algebra. Why must their encodings remain distinguishable? Explain one problem caused by erasing this distinction.
4. Landau removes face `(0,1,3)` while keeping every graph edge. What changes mathematically? A caller recomputes the outer request hash but retains the old carrier anchor: which check must reject it?
5. The runtime description advertises an adapter. Does this configure the graph tutorial executable? Name the two environment-variable families and the evidence missing from the advertisement.
6. Sixty topology lemmas and a clean axiom report are present. State one warranted claim and two unwarranted claims about the native host.

Round 2: Landau wants the numbers

Use the guide's four-vertex carrier and edge order for questions 7–9.

7. Starting with `c=(3,2,2,1,1)` and `f=(0,-2,-2,-1)`, calculate all five edge differences, all residuals, and the residual cost. Explain why this only gives an upper bound until another argument is supplied.
8. Construct a dual circulation that proves a lower bound of 7. Check balance at every vertex, the unit coordinate caps, and its pairing with `c`. Conclude the exact minimum.
9. Calculate both face sums and Gamma2. Explain why spanning both graph cycles with declared faces supports the zero final normal form under the declared solve without implying that the before cochain was already exact.
10. For the cycle claims `(1,1,1)` and penalties `(1,2,2)`, derive the minimum and a minimizing projected cochain. Now change the penalties to `(1,1,1)`

and construct two distinct optimal residuals. What happens to uniqueness?

11. Compare UTF-8 bytes for U+00E9 and U+0065 U+0301. What does NFC do? Explain why the helper's NFC step does not establish RFC 8785 conformance.
12. An attacker edits a returned outcome and recomputes its unkeyed root. Distinguish an internally consistent record, a fresh replay, an authenticated producer and an admitted result. Which of these has the attacker shown?
13. Consider a declared carrier with 512 vertices, 2,048 edges and one face. All dimension limits are satisfied. Evaluate the four cell products for `d0`, `d1`, the augmented matrix, and the closed-system matrix. Is the preallocation fence satisfied? Also explain why a valid timeout field does not terminate a long solve.
14. Which restricted before/candidate wire spellings are permitted: `"0"`, `"-7"`, `"+7"`, `"007"`, `"-0"`, `"1/2"`? Give a reason for every refusal. Do not infer the restricted wire contract from the internal arithmetic type.

Final round: Woland checks the missing obligations

15. **The perfectly rehashed impostor.** You control every field in an exported JSON outcome and can compute every unkeyed digest. Construct a plausible forgery that passes a root-consistency check but does not demonstrate native replay. Identify the independent evidence a consumer would need to defeat your attack. Finally explain why even a genuine replay of this fixture still has refused admission.
16. **The changed challenge.** Begin with cycle claims `(1,1,1)`. A relational CUT proposes zero on only two edges and omits the third. Describe the correct outcome and state behavior. Compare it with a complete all-zero proposal. Explain why accepting the latter cannot establish that an immutable external observation was wrong.
17. **Two wrappers, one oracle.** Two Python wrappers agree on 100 fixtures but call the same calculator. Design a genuinely independent check for the cycle family and one exact primal/dual check for a graph outside that family. State what the checks establish, which input cases they cover, and what source/binary/provenance obligations remain.
18. **The public reproducibility challenge.** Build an evidence-custody plan for another student to reproduce the native four-vertex path from a standalone public deposit. Specify source/dependencies, fixture and anchors, wire profile, launcher/binary identity, toolchain, commands, process limits, results and refusal fields. Define at least five hostile controls, each with a predicted check or refusal. Keep historical results, your fresh run, skipped tests and unexecuted dependencies distinct.

Optional hints

Question	First hint
7	On edge 20, subtract `f(2)` from `f(0)`
8	A unit circulation around 012 has no net vertex divergence
9	The two faces share edge 01 but have different remaining edges
10	Every residual has the same cycle sum; put all mass on a cheapest edge
11	The composed character uses two UTF-8 bytes; the decomposed pair uses three
12, 15	Ask who supplied the expected anchor and who executed the checker
13	Multiply the edge count by vertices plus faces
14	Parsing must return the same spelling when the parsed integer is serialized minimally
16	An omitted value is not an explicit zero
17	An upper bound plus an independently checked matching lower bound is a certificate
18	Include one altered face, one altered candidate and one oversized request

Grading

Each of questions 1–14 is worth 4 points: 2 for the correct result, 1 for the check or derivation, and 1 for the evidence limit. Each final challenge is worth 11 points: 4 for the concrete construction, 4 for checks and expected outcomes, and 3 for custody and scope. Total: 100.

Missing proof or execution is an open obligation, not a fabricated success. Full credit requires preserving the declared problem and recognizing when a different interface, input, or calculator would change it. The marking scheme is instructional; it is not a validated learner assessment.